1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In bladder cancer, high tumor-infiltrating lymphocyte (TIL) density is associated with improved overall survival (OS) only when MSH6 mismatch repair is intact. In MSH6-deficient tumors, high TIL density does not confer the same OS benefit. Secondary hypothesis: among MSH6-intact, TIL-high tumors, positive stool DNA methylation for SDC2 (and/or TFPI2) attenuates the OS benefit, indicating that tumor DNA shedding and associated systemic immune modulation can undermine effective anti-tumor immunity. This three-way interaction (MSH6 status × TIL density × stool SDC2/TFPI2 methylation) persists after adjustment for MSI status and is potentially modulated by Immunoscore as a secondary immune context marker. Mechanistically, intact MSH6 may support neoantigen-driven, effective T cell responses, whereas MSH6 deficiency can promote immune dysfunction (e.g., exhaustion, impaired antigen presentation) whose benefit is blunted in the presence of detectable stool methylation reflecting higher tumor shedding and/or burden.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, SDC2
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/MSI) both shape bladder tumor immunogenicity, yet high TIL density does not uniformly translate into better survival—why do some TIL-rich tumors fail to benefit? 2) Tier 1 paths used: (a) MSH6 (links MMR status and neoantigen load) and (b) SDC2/TFPI2 stool methylation (marks tumor DNA shedding/non■localized invasive phenotype).

SDC2: SDC2); other genes/markers (TFPI2, MSI, CTCs, HDL■C, ER status) are listed as secondary_modifiers for follow-up rather than being placed directly into the mechanistic claim.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TFPI2	Supporting modifier from knowledge graph

Secondary	MSI	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	ER status	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , MSI status, and OS with appropriate covariate adjustment (stage, grade, treatment). Potential limitations include bladder cancer heterogeneity, sample availability for matched tissue/stool analyses, and the need for prospective validation.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TFPI2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.